Fiscal Year:	FY 2022	**Task Last Updated:**	FY 03/10/2023
PI Name:	Burma, Sandeep Ph.D.		
Project Title:	Mechanistic Analysis of Particle Radiation-Induced Carcinogenesis Using Validated Mouse Glioma Models		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:	NOTE: Formerly at University of Texas Southwestern Medical Center at Dallas until fall 2019.		
Project Type:	Flight	**Solicitation / Funding Source:**	2014-15 HERO NNJ14ZSA001N-RADIATION. Appendix D: Ground-Based Studies in Space Radiobiology
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No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
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Flight Program:			
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Key Personnel Changes/Previous PI:	Dr. Bipasha Mukherjee and Dr. Nozomi Tomimatsu are CoInvestigators on the project. Bipasha Mukherjee, PhD, Associate Professor, UT Health San Antonio – has extensive experience in glioblastoma biology and radiation resistance and her expertise will be critical for this project on radiation-induced glioblastoma development. Nozomi Tomimatsu, PhD, Assistant Professor, UT Health San Antonio – has extensive experience in the signaling and repair of radiation-induced DNA damage and her expertise will be critical for this project on radiation-induced glioblastoma development.		
COI Name (Institution):	Tomimatsu, Nozomi Ph.D. (University of Texas Health San Antonio) Mukherjee, Bipasha Ph.D. (University of Texas Health San Antonio)		
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Performance Goal No.:			
Performance Goal Text:			

Task Description:	NOTE: Continuation of "Mechanistic Analysis of Particle Radiation-Induced Carcinogenesis Using Validated Mouse Glioma Model," grant NNX16AD78G, with the same principal investigator, Dr. Sandeep Burma. This is due to PI's move to University of Texas Health Science Center San Antonio in fall 2019. Galactic Cosmic Rays (GCR) and Solar Particle Events (SPEs) are major sources of carcinogenic risk for astronauts on prolonged space missions. Large uncertainties exist in the exact estimation of cancer risks from charged particles due to the paucity of epidemiological data in this area. Furthermore, a mechanistic understanding of genetic changes underlying transformation by particle radiation is necessary for the development of appropriate countermeasures. Thus, experimental animal models that closely mimic the process of carcinogenesis in humans are essential for oncogenic risk assessment. In the course of previously funded research, we characterized two sensitive and complementary mouse models that can be used to elucidate molecular mechanisms underlying the process of particle radiation-induced glioblastoma (GBM) development. GBMs are lethal brain tumors with very dismal prognosis for which radiation is the only known risk factor. GBMs represent the third leading cause of cancer-related death among adults aged 30 – 50 years (the average age of astronauts), the mean survival after diagnosis being only about 14 months. GBM was the first cancer to be analyzed by The Cancer Genome Atlas Network (TCGA), and the key genetic alterations occurring in GBM are now well defined. Based on this information, we utilized transgenic mice with brain-restricted deletions of GBM-relevant tumor suppressors in logical combinations to analyze the process of charged particle-induced carcinogenesis. We identified and characterized two complementary mouse GBM models that would be ideal for studying particle radiation-induced carcinogenesis -- Nestin-Cre; Ink4ab-/-Arf f/f, and Nestin-Cre; p53f/+Ptenf/+. These models exhibit a low frequency of spontaneous GBMs but readily develop brain tumors after exposure to HZE (high energy) particles, which resemble human high grade gliomas in their genetic and molecular signatures. Using these mice, we have clearly demonstrated that high-linear energy transfer (LET) charged particles have a greater transforming potential compared to low-LET radiation, and have characterized the molecular events underlying Fe ion-induced gliomagenesis in these models. This work sets the stage for quantitative studies on radiation quality effects on carcinogenesis for an array of charged particles representing the GCR (as proposed in Aim 1). These charged particles induce complex DNA double-strand breaks (DSBs) the accurate repair of which is critical for preventing transformation. Our models will also allow us to determine which DSB repair pathway represents a major barrier to charged particle-induced gliomagenesis (as proposed in Aim 2). This information would be fundamental to the development of a mechanistic understanding of charged particle-induced carcinogenesis. We will cross the transgenic mice characterized by us with mice harboring brain-specific ablation of non-homologous end joining (NHEJ) or homologous recombination (HR) repair pathways -- Nestin-Cre; Ligase4f/f, and Nestin-Cre; Brca2f/f – that were established in the laboratory of Prof. Peter McKinnon (collaborator on project). Finally, genomic and bioinformatics analyses will be used to define genetic signatures unique to particle-radiation induced glioblastomas and to delineate key pathways that are de-regulated during HZE-induced gliomagenesis (as proposed in Aim 3). We are hopeful that an in-depth mechanistic analysis of charged particle-induced gliomagenesis in these validated mouse models will help us to make significant progress towards cancer risk assessments and development of protective strategies for charged particle exposure.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	As cancer therapy with particles (protons and C ions) is becoming more widely used, our research with heavy ions is also important from a therapeutic standpoint here on Earth. Our examination of the relationship between LET and gliomagenesis could have practical implications for radiotherapy modality choice.
	Charged particles induce more complex DNA damage compared to low LET radiation and several studies show that these particles are more transforming than X- or gamma-rays in vitro. This raises the possibility that HZE ions may pose a significant cancer risk to astronauts. However, large uncertainties exist in the estimation of cancer risks from the space radiation environment. Given the paucity of data from exposed astronauts, the use of animal models in a simulated space radiation environment is a good alternative to estimate these cancer risks. A limited number of animal studies have been carried out to evaluate the carcinogenic effects of charged particles. The most comprehensive of these were the mouse Harderian gland studies that established quantitative relationships between tumor frequency versus charge, LET or dose. However, Harderian gland tumors do not occur in humans as humans lack these glands. A number of recent studies have addressed the question of HZE carcinogenesis using human-relevant tumor models of hematopoietic, intestinal, liver, mammary, lung and brain cancers (our studies). The overarching conclusion of these experiments, including ours, has been that HZE particles are more tumorigenic than low-LET radiation, at least for solid tumors. Some of these studies have found a correlation between radiation quality and transformation with a peak followed by a plateau at higher LETs. These studies set the stage for more accurate estimations of relative biological effectiveness (RBE) for a range of GCR-relevant particles and a quantitative assessment of the relationship between radiation quality and RBE. Glioblastoma (GBM) are lethal brain cancers which can be triggered by even low dose of ionizing radiation (IR); hence, GBM is a good paradigm to study radiation quality effects in the context of carcinogenesis triggered by space radiation. The mouse models of particle radiation-induced glioblastoma that we have validated and characterized with NASA funding are uniquely suited for assessing radiation quality effects and understanding mechanisms underlying particle radiation-driven carcinogenesis. An ideal carcinogenesis model should display a low frequency of spontaneous tumors but be susceptible to radiation-induced tumor development. With support from NASA, we identified and characterized two sensitive (and complementary) mouse models of particle radiation-induced gliomagenesis. To develop these models, we focused on transgenic mice bearing floxed (f) alleles of the most critical, glioma-relevant TSGs (Ink4a, Ink4b, Arf, p53 and/or PTEN) in logical combinations. Targeted TSG combinations were deleted specifically in the mouse brain by crossing these mice to a Nestin-Cre mouse. The Ink/Arf and p53 TSG deletions are generally mutually exclusive in GBM; hence, we used two distinct cohorts of mice – one with deletions of Ink4a, Ink4b and/or Arf, and the other with loss of one or both copies of p53 and/or Pten. All these models (40 mice per genotype) were intra-cranially irradiated with 600 MeV/n Fe ions (0.5 Gy or 1 Gy) and monitored for tumor development. The resulting tumors were subjected to extensive analyses to confirm tumor grade, glial origin, and underlying genomic changes. By this large scale and systematic approach, we identified two complimentary mouse models that would be ideal for estimating cancer risks from particle radiation (relative to low LET radiation) - a) Nestin-Cre; Ink4ab-/-Arf f/f and b) Nestin-Cre; p53f/+Ptenf/+. The radiogenic gliomas in both mouse models exhibited histopathological characteristics of human grade III/IV disease and were driven by GBM-relevant oncogenic changes such as Met amplification (see below). Using these mouse models, we are evaluating the gliomagenic properties of an array of charged particles with a range of LETs (0.53 - 173.6 KeV/µm) using both model systems. We intra-cranially irradiated cohorts of

Task Progress:	Nestin-Cre;Ink4ab-/-;Arf f/f mice with 0.5 Gy of X-rays, H, C, Ne, Si, Ti and Fe, and Nestin-Cre;p53f/+;PTENf/+ with 0.5 Gy of X-rays, H, C, Si, and Fe and monitored for glioma formation. We were able to demonstrate tumor formation following exposure to 0.5 Gy of these ions. All observed radiogenic tumors, regardless of model or radiation quality, phenocopy the histopathological features of high-grade human gliomas, including aggressive, infiltrative growth, hypercellularity, high mitotic index, pseudopalisading necrosis, and nuclear atypia. Importantly, an increase GBM frequency was observed with increasing atomic number with a maximum transformation frequency observed with Si irradiation. The same trend was observed regardless of the mouse model, underscoring the universality of the link between LET and gliomagenesis. These results demonstrate a linear relationship between LET of charged particles and GBM development that peaks around Si (LET=79 KeV/µm). We also used the simplified GCR (H, C, O, Si, and Fe) simulations on both mouse models. In preliminary studies we observed induction of tumor development in both models after irradiation with GCRsim. The follow up GCRsim experiments proposed for 2020 had to be deferred due to the COVID crisis. We are planning on carrying out these irradiations later in 2021. Next, we carried out onco-genomic analyses of the radiogenic gliomas to identify tumor-cell intrinsic genomic changes triggered by heavy ion exposure that promote tumor development. We first analyzed tumors by array CGH with the goal of identifying common regions of deletion or amplification that would presumably harbor changes critical for tumorigenesis, thereby distinguishing between “driver” and “passenger” mutations. Genomic DNA derived from tumors were compared to DNA from normal mouse brain of the corresponding genotype by array CGH. By these analyses we found that amplification of the receptor tyrosine kinase (RTK) Met occurred very frequently in both mouse models. Interestingly, we found that Met signaling lead to the expression of Sox2 which, in turn, supported a cancer stem cell phenotype in the radiogenic gliomas. These observations are significant as they indicate that such radiation-induced tumors might be more resistant to therapy (as cancer stem cells are understood to drive therapeutic resistance in tumors). We have also carried out preliminary whole genome sequencing studies to determine whether radiation quality could contribute to distinct mutational signatures in gliomas derived from different LETs. We found that genome-wide there was a qualitative difference in the size of deletions between electromagnetic and particulate radiation-induced GBMs. We also hypothesized that tumor-cell extrinsic changes to the microenvironment would be a significant contributor to heavy-ion induced gliomagenesis. Given the frequent amplification of the RTK Met in both tumor models, we wanted to examine whether secretion of the Met ligand – HGF – by the irradiated stromal cells might drive selection of nascent tumor cells with Met amplification. For these studies we resorted to X-ray irradiation as the Covid crisis prevented us from utilizing the beams at NASA Space Radiation Laboratory (NSRL) for the entirety of 2020. In these preliminary studies, we found that IR can trigger senescence of astrocytes in the brain and that these senescent astrocytes secrete a number of cytokines, growth factors, and metalloproteases which make up the senescence -associated secretory phenotype (SASP) with known tumor promoting roles. Very interestingly, we observed upregulation of HGF (a bona fide SASP factor) in the irradiated brain as well as secretion of HGF by senescent astrocytes which resulted in Met activation in the glioma cells and increased growth and invasiveness of tumor cells both in vitro and in vivo. These findings provide us with a mechanistic framework to understand how space radiation-induced changes to both the incipient tumor cell and the tumor microenvironment drives tumor initiation and progression. These findings may also help us develop therapeutic approaches targeting both the pre-cancerous cells and senescent stromal cells with senolytic drugs as a radiation countermeasure approach. With greater accessibility to the beams in 2021 (after Covid-related shutdowns), we embarked on recapitulating our previous senescence-related findings with HZE particles (including GCRsim) and carrying out studies aimed at evaluating the effectiveness of senolytic drugs in eliminating senescence in the irradiated brain and mitigating the detrimental effects of space radiation. As a first step, we irradiated primary human astrocytes, preadipocytes, and fibroblasts with protons, Si or Fe ions or with gamma rays for comparison. By staining for senescence associated-beta-galactosidase activity and by evaluating the upregulation of p21 or p16 senescence-enforcing genes, we determined that charged particles, especially Si, were significantly more potent than gamma rays in triggering senescence in these primary cell types. Next, we carried out bulk RNA-seq analyses of irradiated dermal fibroblasts and found that key SASP genes were highly upregulated with space radiation compared to gamma radiation. We also saw transcriptional heterogeneity between different treatment groups. Moreover, charged particle irradiated samples have a higher pathogenic index when compared to gamma radiation, in particular those cells exposed to Si ions. We also carried out preliminary single-cell RNA sequencing analyses of irradiated (senescent) dermal fibroblasts. Interestingly, we found a sub-cluster of cells that were highly enriched in cancer-relevant genes only in cells irradiated with Si. These findings are tantalizing because we find that Si is the most transforming out of all the ions we have tested. Further analyses are needed to understand the link between Si-induced senescence and cellular transformation. Results from these analyses might guide the future development of senolytics as radiation countermeasures.
Bibliography Type:	Description: (Last Updated: 12/09/2025)
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